

Perpetual Committee Selection

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ABSTRACT

We study temporal committee elections, where a fixed-size committee is selected at each timestep and voters' approval preferences may evolve over time. The goal is to balance efficiency and fairness across repeated collective decisions.

We adopt an egalitarian notion of fairness as a hard constraint: each agent is guaranteed a minimum level of representation over the entire time horizon. Among all committee sequences that satisfy this fairness requirement, we seek to maximize utilitarian welfare, defined as the total number of representation events across all agents and timesteps.

We propose a integer linear programming formulation that exactly captures this objective and computes a welfare-optimal sequence of committees subject to the given representation guarantees. Our framework provides a clean and tractable approach to temporal committee selection with guaranteed representation.

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1 INTRODUCTION

Decision-making processes in multiagent systems often involve repeated choices made over time under evolving preferences. Examples include recurring committee selections, allocation of public resources, and governance in online communities. In such settings, it is insufficient to evaluate outcomes in isolation, as agents' satisfaction accumulates across rounds.

Temporal committee elections capture this setting by requiring the selection of a fixed-size committee at each timestep, while allowing agents' approval preferences to vary over time. This model subsumes temporal single-winner elections (when $k = 1$) and static approval-based multiwinner voting (when $T = 1$) as special cases, while focusing on the more general setting of dynamic temporal committee selection and introducing new challenges at the intersection of efficiency and fairness.

Fairness in repeated committee selection can be approached from different perspectives. One natural requirement is to ensure that each agent receives a minimum cumulative level of representation over the time horizon. Such individual-level guarantees prevent systematic under-representation while remaining compatible with optimization-based formulations.

While fairness constraints provide representation guarantees, they may conflict with welfare maximization. In particular, outcomes that maximize utilitarian welfare may leave some agents

systematically under-represented, whereas enforcing fairness constraints may reduce total welfare. Understanding this trade-off is central in temporal committee selection.

In this paper, we study utilitarian and egalitarian welfare objectives in temporal committee elections, their computational properties, and their interaction with individual minimum representation guarantees over time. Our goal is to understand the trade-offs between welfare and fairness and to characterize the extent to which fairness can be enforced without incurring excessive welfare loss.

2 PROBLEM DEFINITION

Let $N = \{1, \dots, n\}$ be a set of agents, P a set of candidates, and $\ell \in \mathbb{N}$ the number of timesteps. At each timestep $t \in [\ell]$, each agent $i \in N$ submits an approval set $S_{i,t} \subseteq P$. A fixed committee size k is given.

An outcome is a sequence of committees $C = (C_1, \dots, C_\ell)$ such that $C_t \subseteq P$ and $|C_t| = k$ for all $t \in [\ell]$. Agent i is said to be *represented* at timestep t if $C_t \cap S_{i,t} \neq \emptyset$.

The goal is to compute a sequence of committees that maximizes utilitarian welfare aggregated over all timesteps while satisfying egalitarian fairness constraints that guarantee each agent a minimum level of representation over the time horizon.

2.1 Computational Complexity

THEOREM 2.1. *The temporal committee selection problem with utilitarian welfare maximization subject to egalitarian fairness constraints is NP-hard.*

PROOF. We prove NP-hardness via a polynomial-time reduction from the Maximum Coverage problem, which is known to be NP-hard.

An instance of Maximum Coverage consists of a universe $U = e_1, \dots, e_n$ of elements, a collection of subsets $\mathcal{S} = S_1, \dots, S_m$ with $S_j \subseteq U$, and an integer k . The goal is to select at most k subsets whose union covers the maximum number of elements.

Given an instance of Maximum Coverage, we construct an instance of our committee selection problem as follows. For each element $e_i \in U$, we create an agent i . For each subset $S_j \in \mathcal{S}$, we create a candidate j . We consider a single timestep $T = 1$ and set the committee size to k .

Agent i approves candidate j if and only if $e_i \in S_j$. The utilitarian welfare of a committee equals the number of agents who are represented, i.e., the number of elements covered by the chosen subsets. We set the egalitarian fairness bound to zero, making the fairness constraint vacuous.

Under this construction, any feasible committee of size k corresponds exactly to a choice of k subsets in the Maximum Coverage instance, and the utilitarian welfare equals the number of covered elements. Therefore, an optimal solution to the constructed committee selection instance yields an optimal solution to the Maximum Coverage problem.

The reduction clearly runs in polynomial time. Since Maximum Coverage is NP-hard, it follows that our committee selection problem is NP-hard. $\square$

COROLLARY 2.2. *Unless $P = NP$, the temporal committee selection problem does not admit a polynomial-time exact algorithm. This justifies the use of integer linear programming formulations for computing optimal solutions.*

3 WELFARE FUNCTIONS

We consider two classical social welfare objectives.

Utilitarian Welfare. The utilitarian welfare of an outcome C is defined as

$$\text{UTIL}(C) = \sum_{i \in N} u_i(C),$$

where

$$u_i(C) = |\{t \in [\ell] \mid C_t \cap S_{i,t} \neq \emptyset\}|$$

denotes the number of timesteps at which agent i is represented.

Egalitarian Welfare. The egalitarian welfare of an outcome C is defined as

$$\text{EGAL}(C) = \min_{i \in N} u_i(C),$$

capturing the satisfaction of the least-represented agent.

Utilitarian welfare emphasizes overall efficiency, while egalitarian welfare prioritizes fairness by protecting agents from systematic under-representation. These objectives are generally incompatible and exhibit different computational properties in temporal committee elections.

4 WELFARE–FAIRNESS TRADEOFF

Temporal committee selection involves a natural tension between efficiency and fairness. The utilitarian objective seeks to maximize the total number of representation events across all agents and timesteps, while fairness requires preventing persistent under-representation of any agent over time.

Let $C = (C_1, \dots, C_\ell)$ denote a sequence of committees. For each agent $i \in N$, define

$$u_i(C) = |\{t \in [\ell] \mid C_t \cap S_{i,t} \neq \emptyset\}|$$

as the number of rounds in which agent i is represented.

Cumulative Egalitarian Fairness. To measure fairness dynamically, we consider the minimum cumulative representation achieved by any agent:

$$\lambda(C) = \min_{i \in N} u_i(C).$$

This quantity reflects the satisfaction of the least-represented agent over the entire time horizon.

Minimum Fairness Guarantee. We introduce a fairness threshold $F \geq 0$ representing the minimum number of times every agent must be represented over the horizon. A committee sequence is *F-fair* if:

$$\lambda(C) \geq F.$$

Any sequence that does not satisfy this threshold is considered infeasible, regardless of its utilitarian welfare. Among all *F-fair* sequences, we select the one maximizing utilitarian welfare, breaking ties by maximizing λ .

Tradeoff Principle. Rather than imposing a fixed fairness bound, we adopt a lexicographic objective over *F-fair* solutions:

- (1) maximize utilitarian welfare, and
- (2) among all welfare-maximizing outcomes, maximize the minimum cumulative representation.

This approach ensures that fairness is improved whenever possible without sacrificing global efficiency. The dynamic notion of fairness naturally captures cumulative representation guarantees while avoiding infeasibility issues that may arise from imposing exogenous fairness thresholds.

5 INTEGER LINEAR PROGRAMMING FORMULATION

We now present an integer linear programming formulation that computes a welfare-optimal sequence of committees while incorporating cumulative egalitarian fairness.

5.1 Variables

For each candidate $p \in P$ and timestep $t \in [\ell]$,

$$x_{p,t} \in \{0, 1\}$$

indicates whether candidate p is selected at timestep t .

For each agent $i \in N$ and timestep $t \in [\ell]$,

$$y_{i,t} \in \{0, 1\}$$

indicates whether agent i is represented at timestep t .

To track cumulative representation, we introduce variables

$$z_{i,t} \geq 0,$$

denoting the total number of times agent i has been represented up to and including timestep t .

Finally, we introduce a variable

$$\lambda \geq 0$$

representing the minimum cumulative representation achieved by any agent over the entire time horizon.

A fairness threshold $F \geq 0$ is a problem parameter representing the minimum guaranteed representation every agent must receive over the time horizon.

5.2 Objective

We adopt a lexicographic objective that prioritizes utilitarian welfare and uses egalitarian fairness as a tie-breaking criterion:

$$\max \sum_{i \in N} \sum_{t=1}^{\ell} y_{i,t} + \varepsilon \lambda,$$

where $\varepsilon > 0$ is a sufficiently small constant satisfying

$$\varepsilon < \frac{1}{|N| \cdot \ell + 1},$$

ensuring that the utilitarian term strictly dominates the fairness term. Specifically, no improvement in λ can compensate for a unit loss in utilitarian welfare, thereby implementing a true lexicographic optimization: the formulation first maximizes total representation events across all agents and timesteps, and among all solutions achieving the same utilitarian welfare, selects the one maximizing the minimum cumulative representation λ .

REMARK 1. Since $y_{i,t} \in \{0, 1\}$, the maximum possible utilitarian welfare is $|N| \cdot \ell$. For any two solutions differing by one unit in utilitarian welfare, the gap is at least 1. Since $\lambda \leq \ell$, the fairness bonus satisfies $\varepsilon\lambda < \varepsilon \cdot \ell < \frac{\ell}{|N| \cdot \ell + 1} < 1$, so fairness can never outweigh a utilitarian difference. Hence the objective correctly implements lexicographic ordering.

5.3 Constraints

Committee Size. Each timestep selects exactly k candidates:

$$\sum_{p \in P} x_{p,t} = k \quad \forall t \in [\ell].$$

Representation Consistency. An agent is represented if and only if at least one approved candidate is selected:

$$y_{i,t} \leq \sum_{p \in S_{i,t}} x_{p,t} \quad \forall i, t,$$

$$\sum_{p \in S_{i,t}} x_{p,t} \leq |S_{i,t}| y_{i,t} \quad \forall i, t.$$

Cumulative Representation Dynamics. Cumulative representation evolves over time:

$$z_{i,1} = y_{i,1} \quad \forall i \in N,$$

$$z_{i,t} = z_{i,t-1} + y_{i,t} \quad \forall i \in N, t = 2, \dots, \ell.$$

Egalitarian Fairness. The variable λ captures the minimum cumulative representation:

$$\lambda \leq z_{i,\ell} \quad \forall i \in N.$$

Variable Domains.

$$x_{p,t}, y_{i,t} \in \{0, 1\}, \quad z_{i,t} \geq 0, \quad \lambda \geq 0.$$

Minimum Fairness Threshold. Every agent must be represented at least F times:

$$\lambda \geq F.$$

This hard constraint eliminates all committee sequences that fail to guarantee each agent a minimum of F representation events, regardless of their utilitarian welfare.

REMARK 2. A natural default is $F = \lfloor kT/n \rfloor$, representing the proportionally fair share of representation each agent would receive if committee seats were distributed evenly. Setting $F = 0$ recovers pure utilitarian optimization. Setting $F = \lambda_{\max}$, where $\lambda_{\max}$ is the maximum achievable minimum representation, enforces the strongest possible fairness guarantee. In practice, F should satisfy $0 \leq F \leq \lambda_{\max}$ to ensure feasibility; if $F > \lambda_{\max}$, the formulation becomes infeasible.

5.4 Interpretation

The formulation optimizes over all feasible committee sequences implicitly through the binary decision variables. The hard constraint $\lambda \geq F$ ensures that only sequences guaranteeing every agent at least F representation events are considered; sequences failing this threshold are eliminated regardless of their utilitarian welfare. Among feasible sequences, the primary objective maximizes total representation events, while the fairness variable λ ensures that, among equally efficient solutions, preference is given to those improving representation of the least-served agent.